# Monopoly

EC 111 Principles of Microeconomics

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# Reading

- These slides are not a substitute for reading Mankiw, Chapter 16, Monopoly.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 16, "Monopoly".

# What you must know from this chapter

1. Definition of a monopoly.
2. How can a monopoly arise?
3. Difference between P and MR.
4. Graphical representation of profits/losses.
5. Barriers to entry.
6. Comparison to perfect competition.
7. Natural monopoly.

Part 1

# Monopoly and Market Power

# Monopoly

**Definition: Monopoly**
A firm that is the sole seller of a product without close substitutes.

**Definition: Market power**
The ability of a firm to influence the market price of what it sells.

- A competitive firm is a price taker. A monopoly is a price maker.

# Monopoly: examples

- **The idea:** a town's water network; a drug company selling a medicine while its patent runs. Each is the only seller, and buyers have no close substitute.
- **Found by US courts:** in cases brought by the Department of Justice, judges ruled that Google is a monopolist in online search (2024) and in online advertising technology (2025). Google has appealed the search ruling.
- **The Federal Trade Commission** also brings monopoly cases. In 2025 a court ruled that Meta (Facebook, Instagram, WhatsApp) is not a monopoly. The FTC's case against Amazon goes to trial in March 2027.

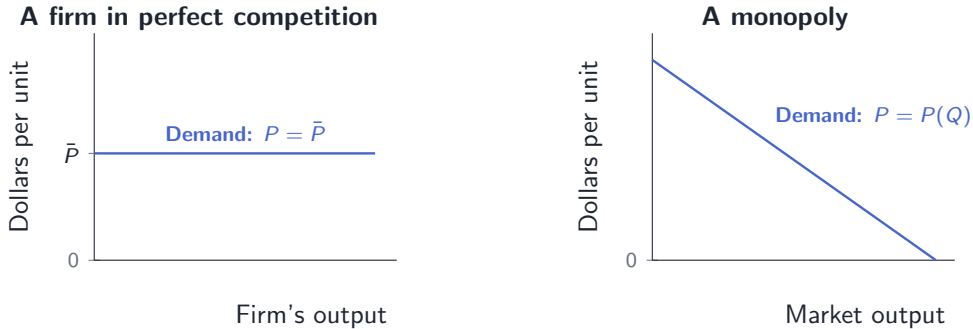
# A price maker faces the market demand curve

- A competitive firm sells as many units as it likes at the market price: price is parametric, $P = \bar{P}$.
- A monopoly is the whole market. It faces the market demand curve, so its price depends on its output: $P = P(Q)$.
- Demand slopes down. If the monopoly wants to sell more, it must lower its price.
- It chooses $Q$; the demand curve then gives the highest price buyers will pay, $P(Q)$.

**Finding**

A monopoly cannot charge any price and sell any quantity. It can only choose a point on its demand curve.

# Competitive firm versus monopoly




**Finding**

A competitive firm faces a horizontal demand line at $\bar{P}$. A monopoly faces the downward-sloping market demand curve, so selling more means a lower price.

# Example: a printer cartridge seller

One firm is the only seller of a printer cartridge. Its demand curve:

$$P(Q) = 32 - 2Q.$$

Each extra cartridge sold lowers the price by \$2.

To sell 6 cartridges, the most it can charge is \$20. To charge \$24, it can sell only 4.

| $Q$ | $P(Q) = 32 - 2Q$ |
|---|---|
| 2 | \$28 |
| 4 | \$24 |
| 6 | \$20 |
| 8 | \$16 |

# Part 1 summary

- A monopoly is the sole seller of a product without close substitutes.
- It is a price maker: it faces the market demand curve, $P = P(Q)$.
- To sell more it must lower its price. It can only choose points on its demand curve.

# Your turn: Check your understanding

The cartridge seller's demand is $P(Q) = 32 - 2Q$.

1. At 4 cartridges, the price is:
   A. \$16  B. \$24  C. \$28
2. To charge \$20, it can sell:
   A. 4  B. 6  C. 10
3. A monopoly can charge any price and still sell any quantity.
   A. True  B. False

| $Q$ | $P(Q)$ |
|---|---|
| 2 | \$28 |
| 4 | \$24 |
| 6 | \$20 |
| 8 | \$16 |

# Solution: Check your understanding

1. **B.** $P(4) = 32 - 2 \times 4 = \$24$.
2. **B.** $20 = 32 - 2Q$, so $Q = 6$.
3. **B.** False: it can only choose points on its demand curve.

| $Q$ | $P(Q)$ |
|---|---|
| 2 | \$28 |
| 4 | \$24 |
| 6 | \$20 |
| 8 | \$16 |

# Check: Part 1

Monopoly firms face

A. downward-sloping demand curves, so they can sell as much output as they desire at the market price.

B. downward-sloping demand curves, so they can sell only the specific price-quantity combinations that lie on the demand curve.

C. horizontal demand curves, so they can sell as much output as they desire at the market price.

D. horizontal demand curves, so they can sell only a limited quantity of output at each price.

# Check: Part 1 – answer

Monopoly firms face

A downward-sloping demand curves, so they can sell as much output as they desire at the market price.

**B downward-sloping demand curves, so they can sell only the specific price-quantity combinations that lie on the demand curve.**

C horizontal demand curves, so they can sell as much output as they desire at the market price.

D horizontal demand curves, so they can sell only a limited quantity of output at each price.

**Finding**

A monopoly is the whole market, so it faces the downward-sloping market demand curve. Each output has one highest price, $P(Q)$. Horizontal demand describes a competitive firm.

Part 2

# Barriers to Entry and Natural Monopoly

# Barriers to entry

**Definition: Barrier to entry**

Anything that stops other firms entering a market and competing.

Without entry barriers, profit attracts entry and the monopoly disappears. Monopolies arise from three sources:

1. **A key resource** is owned by one firm (a unique mineral deposit).
2. **The government** gives one firm the exclusive right to sell (a patent or a copyright).
3. **Costs:** one firm can supply the whole market more cheaply than two or more firms. This is a natural monopoly.

# Barriers created by government

- **Patent:** the inventor alone may make and sell the invention. In the US a patent lasts 20 years from the date the application is filed.
- **Copyright:** the author alone may copy and sell a book, song, film or piece of software. In the US it lasts for the author's life plus 70 years.
- **When they expire,** the barrier is removed and other firms can enter. When a drug's patent ends, generic versions enter and the price falls.

# Natural monopoly

**Definition: Natural monopoly**

A market in which one firm can supply the whole market at a lower average total cost than two or more firms could.

**Example:** the water supply network in a town. Laying the pipes is expensive. Once they are laid, connecting one more home costs little. One network serves the whole town more cheaply than two networks side by side.

# Why one firm is cheaper: falling average total cost

- A natural monopoly arises when fixed cost FC is large and marginal cost $c$ is small and constant: $\text{TC} = \text{FC} + c \times Q$.
- Then $\text{ATC} = \dfrac{\text{FC}}{Q} + c$. As $Q$ rises, FC is spread over more units, so ATC falls.

**Finding**

If ATC falls over the whole market, splitting output between two firms raises ATC: each firm pays the fixed cost.

# Example: a town water network

Laying the pipes costs \$120 (FC). Each water connection to a home costs \$3 (MC):

$$\text{TC} = 120 + 3Q, \qquad \text{ATC} = \frac{120}{Q} + 3.$$

ATC falls as water connections rise: \$9 at 20, \$5 at 60.

ATC is always above MC, which is \$3.

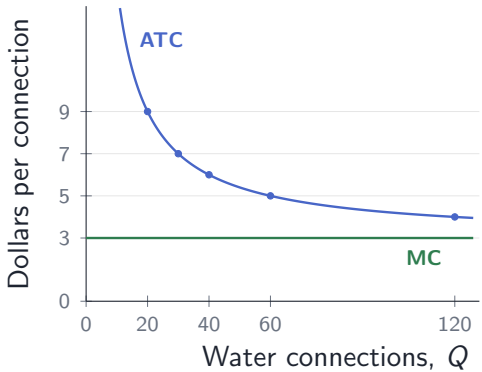
| $Q$ | TC | ATC |
|---|---|---|
| 20 | \$180 | \$9 |
| 30 | \$210 | \$7 |
| 40 | \$240 | \$6 |
| 60 | \$300 | \$5 |

# Natural monopoly on the graph

$Q$ is the number of water connections to homes. ATC $= 120/Q + 3$ falls over the whole range.

MC is flat at \$3, below ATC at every output.

The more homes one water network connects, the lower its cost per connection.



# One network or two?

The town needs 60 water connections.

- One network: $\text{TC} = 120 + 3 \times 60 = \$300$. $\text{ATC} = \$5$.
- Two water networks of 30 each: $2 \times (120 + 3 \times 30) = 2 \times 210 = \$420$. $\text{ATC} = \$7$.
- Compare total costs, two networks against one: $\$420 - \$300 = \$120$ more. The extra \$120 is the second fixed cost.

**Finding**

In a natural monopoly, one firm supplies the market at the lowest cost. A second firm would duplicate the fixed cost.

# Part 2 summary

- Barriers to entry are the cause of monopoly: without them, entry competes profit away.
- Three sources: a key resource, government-created rights, and costs.
- Natural monopoly: large FC and low MC make ATC fall as output rises, so one firm supplies the market most cheaply.

# Your turn: Check your understanding

The town water network: $\text{TC} = 120 + 3Q$.

1. ATC at 40 connections is:

   A. \$3  B. \$6  C. \$240

2. Serving 40 connections with two networks of 20 instead of one costs extra:

   A. \$0  B. \$60  C. \$120

3. A patent creates a barrier to entry.

   A. True  B. False

| $Q$ | TC | ATC |
|---|---|---|
| 20 | \$180 | \$9 |
| 40 | \$240 | \$6 |

# Solution: Check your understanding

1. **B.** ATC $= 120/40 + 3 = 3 + 3 = \$6$.
2. **C.** Two networks: $2 \times 180 = \$360$; one: \$240. Extra: $360 - 240 = \$120$.
3. **A.** True: the government gives one firm the exclusive right to sell.

| $Q$ | TC | ATC |
|---|---|---|
| 20 | \$180 | \$9 |
| 40 | \$240 | \$6 |

# Check: Part 2

When an industry is a natural monopoly,

A it is characterized by constant returns to scale.

B it is characterized by diseconomies of scale.

C a larger number of firms may lead to a lower average total cost.

D a larger number of firms will lead to a higher average total cost.

# Check: Part 2 – answer

When an industry is a natural monopoly,

A it is characterized by constant returns to scale.

B it is characterized by diseconomies of scale.

C a larger number of firms may lead to a lower average total cost.

**D a larger number of firms will lead to a higher average total cost.**

**Finding**

ATC falls as one firm's output rises. Splitting output among more firms means each pays the fixed cost and produces less, so ATC is higher: the network's \$5 becomes \$7 with two firms.

Part 3

# Total, Average and Marginal Revenue

# Total, average and marginal revenue

**Definition: Total revenue**

$\text{TR}(Q) = P(Q) \times Q$: the price depends on output.

**Definition: Average revenue**

$\text{AR} = \text{TR}/Q = P(Q)$: revenue per unit is the price, as in competition.

**Definition: Marginal revenue**

$\text{MR} = \Delta\text{TR}/\Delta Q$: the extra revenue from an extra unit. $\Delta Q = 1$.

# Selling more changes revenue in two ways

Total revenue is $\text{TR} = P \times Q$. Raise output by $\Delta Q$. To sell the extra units, the price must change by $\Delta P$ (a fall, so $\Delta P < 0$).

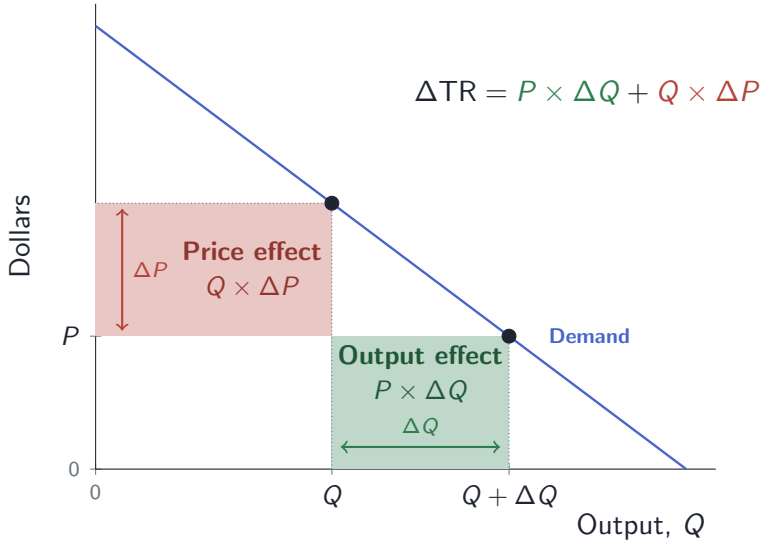
- **Output effect:** the extra $\Delta Q$ units sell at the new price $P$: revenue rises by $P \times \Delta Q$.
- **Price effect:** the (old) $Q$ units already sold now sell at a lower price: revenue changes by $Q \times \Delta P$, which is negative ($Q$ is the number of units sold before the change).
- Together:

$$\Delta \text{TR} = P \times \Delta Q + Q \times \Delta P.$$

**Finding**

$P$ is the new price; $Q$ is the number of units already sold.

# The fundamental trade-off



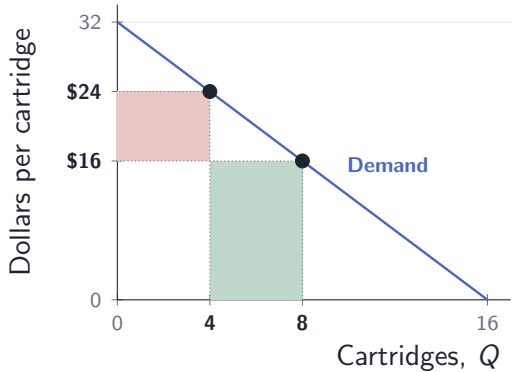
# The two effects on the graph

The cartridge seller goes from 4 to 8 units. The price falls from \$24 to \$16: $\Delta Q = 4$, $\Delta P = -8$.

**Output effect** (green): 4 more units at \$16: $16 \times 4 = +\$64$.

**Price effect** (red): the first 4 units lose \$8 each: $4 \times (-8) = -\$32$.

$\Delta \text{TR} = 64 - 32 = \$32$. Check: TR goes from \$96 to \$128.




Source of the numbers: $P = 32 - 2Q$, so $P(4) = 32 - 8 = \$24$ and $P(8) = 32 - 16 = \$16$; $\text{TR} = P \times Q$: $24 \times 4 = \$96$, $16 \times 8 = \$128$.

# From the change in revenue to marginal revenue

- Start from the two effects:

$$\Delta \text{TR} = P \times \Delta Q + Q \times \Delta P.$$

- Divide both sides by $\Delta Q$:

$$\frac{\Delta \text{TR}}{\Delta Q} = P \times \frac{\Delta Q}{\Delta Q} + Q \times \frac{\Delta P}{\Delta Q}.$$

- $\Delta Q / \Delta Q = 1$ and $\Delta \text{TR} / \Delta Q = \text{MR}$, so

$$\text{MR} = P + Q \times \frac{\Delta P}{\Delta Q}.$$

# Marginal revenue is below price (= average revenue)

$$\text{MR} = P + Q \times \frac{\Delta P}{\Delta Q}$$

- Demand slopes down, so $\Delta P/\Delta Q < 0$. So the last term is negative.
- Write it as $-k$, where $k > 0$. Then $\text{MR} = P - k$, so

$$P = \text{MR} + k.$$

- Whatever MR is, $P$ is bigger by $k$. So $P > \text{MR}$.
- $\text{AR} = P$, so also $\text{AR} > \text{MR}$.

# MR for a straight-line demand curve

Demand is a straight line: $P = a - bQ$, with $a > 0$ and $b > 0$. Each extra unit lowers the price by $b$, so $\Delta P / \Delta Q = -b$.

$$
\begin{aligned}
\text{MR} &= P + Q \times \frac{\Delta P}{\Delta Q} \\
&= (a - bQ) + Q \times \frac{\Delta P}{\Delta Q} && \text{replace } P \text{ with } a - bQ \\
&= (a - bQ) + Q \times (-b) && \text{replace } \Delta P / \Delta Q \text{ with } -b \\
&= a - bQ - bQ && Q \times (-b) = -bQ \\
&= a - 2bQ && -bQ - bQ = -2bQ
\end{aligned}
$$

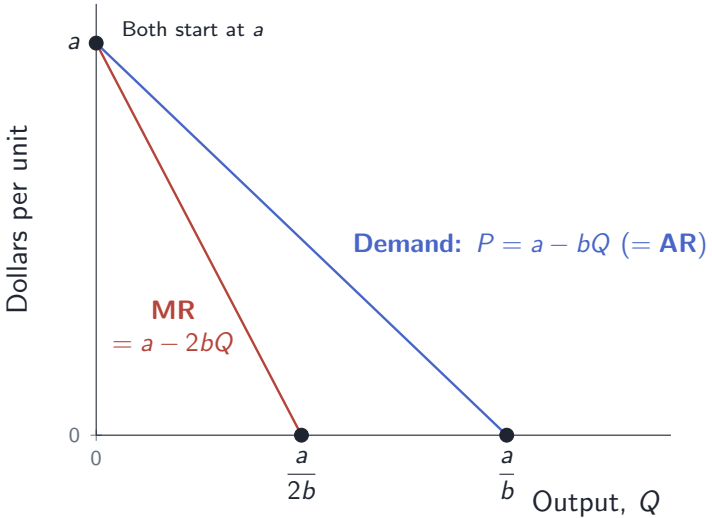
# MR is twice as steep as demand

- Demand: $P = a - bQ$. Each extra unit lowers the price by $b$.
- Marginal revenue: $\text{MR} = a - 2bQ$. Each extra unit lowers MR by $2b$.
- Both lines start at $a$ on the vertical axis.

**Finding**

For every straight-line demand curve, the MR line starts at the same point and is twice as steep. This is a general result, not a feature of one example.

# Demand (AR) and marginal revenue



# Example: the cartridge seller's revenue

| $Q$ | $P(Q)$ | TR | MR |
|---|---|---|---|
| 1 | \$30 | \$30 | \$30 |
| 2 | \$28 | \$56 | \$26 |
| 3 | \$26 | \$78 | \$22 |
| 4 | \$24 | \$96 | \$18 |
| 5 | \$22 | \$110 | \$14 |
| 6 | \$20 | \$120 | \$10 |
| 7 | \$18 | \$126 | \$6 |
| 8 | \$16 | \$128 | \$2 |
| 9 | \$14 | \$126 | −\$2 |

$P(Q) = 32 - 2Q$; $\text{TR} = P(Q) \times Q$; MR in row $Q$ is $\text{TR}(Q) - \text{TR}(Q-1)$.

MR is below the price for every unit after the first.

The 9th unit has negative MR: TR falls from \$128 to \$126.

The price falls \$2 per unit; MR falls \$4 per unit: twice as steep.

# The two effects in the table

Cartridges: $\Delta P/\Delta Q = -2$. Use $\text{MR} = P + Q \times \Delta P/\Delta Q$, with $P$ the new price and $Q$ the units already sold.

- **6th unit:** $\text{MR} = 20 + 5 \times (-2) = 20 - 10 = \$10$.
- **9th unit:** $\text{MR} = 14 + 8 \times (-2) = 14 - 16 = -\$2$.

**Finding**

When the price effect is larger than the output effect, MR is negative: selling more lowers total revenue.

# Part 3 summary: revenue

- Total revenue: $\text{TR}(Q) = P(Q) \times Q$.
- Average revenue equals price: $\text{AR} = P(Q)$.
- Selling more changes revenue in two ways: $\Delta\text{TR} = P \times \Delta Q + Q \times \Delta P$.
- The output effect, $P \times \Delta Q$, raises revenue.
- The price effect, $Q \times \Delta P$, lowers revenue.
- Marginal revenue: $\text{MR} = P + Q \times \Delta P / \Delta Q$.

# Part 3 summary: marginal revenue

- For a monopoly, demand slopes down, so $P > \text{MR}$.
- Because $\text{AR} = P$, also $\text{AR} > \text{MR}$.
- For a straight-line demand curve, $\text{MR} = a - 2bQ$: it starts at the same point and is twice as steep.
- MR can be negative. Then selling more lowers total revenue.
- Compared with competition: a competitive firm's price does not change when it sells more, so $\text{MR} = \bar{P}$.

# Your turn: Check your understanding

The cartridge seller, $P(Q) = 32 - 2Q$.

1. TR at 4 cartridges is:
   A. \$24  B. \$78  C. \$96
2. MR of the 4th cartridge is:
   A. \$18  B. \$24  C. \$96
3. For the 4th cartridge, the price effect changes revenue by:
   A. $-\$2$  B. $-\$6$  C. $-\$24$

| $Q$ | $P(Q)$ | TR |
|---|---|---|
| 3 | \$26 | \$78 |
| 4 | \$24 | \$96 |

# Solution: Check your understanding

1. **C.** $\text{TR} = 24 \times 4 = \$96$.
2. **A.** $\text{MR} = 96 - 78 = \$18$.
3. **B.** $Q \times \Delta P = 3 \times (-2) = -\$6$. Check: $24 - 6 = \$18$.

| $Q$ | $P(Q)$ | TR |
|---|---|---|
| 3 | \$26 | \$78 |
| 4 | \$24 | \$96 |

# Check: Part 3

If a profit-maximizing monopolist faces a downward-sloping market demand curve, its

A average revenue is less than the price of the product.

B average revenue is less than marginal revenue.

C marginal revenue is less than the price of the product.

D marginal revenue is greater than the price of the product.

# Check: Part 3 – answer

If a profit-maximizing monopolist faces a downward-sloping market demand curve, its

A average revenue is less than the price of the product.

B average revenue is less than marginal revenue.

**C marginal revenue is less than the price of the product.**

D marginal revenue is greater than the price of the product.

**Finding**

To sell one more unit, the monopoly lowers the price on all its units. The price effect makes MR $< P$. Average revenue always equals the price, so A and B are wrong.

Part 4

# Profit Maximization and Profit or Loss

# The rule carries over from competition

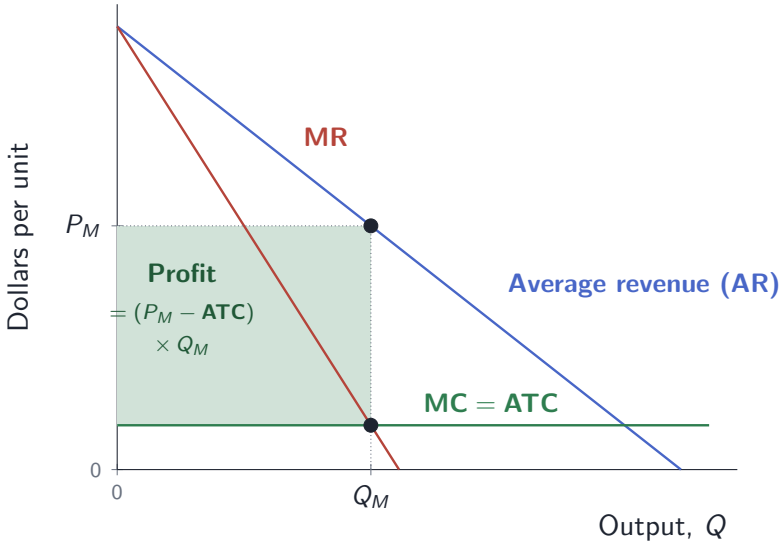
- Profit depends on output: $\pi(Q) = \text{TR}(Q) - \text{TC}(Q)$.
- One more unit changes profit by $\Delta\pi = \text{MR} - \text{MC}$.
- If $\text{MR} > \text{MC}$, increase output. If $\text{MC} > \text{MR}$, cut output.
- Profit is maximized where $\text{MR} = \text{MC}$.

**Finding**

The profit-maximizing rule is $\text{MR} = \text{MC}$. For one more unit:

$$\text{MR} = \text{MC} \;\Rightarrow\; \text{MR} - \text{MC} = 0 \;\Rightarrow\; \Delta\pi = \Delta\text{TR} - \Delta\text{TC} = 0.$$

# Optimal output, price and profit



# Example: the cartridge seller's optimal output

| $Q$ | $P(Q)$ | TR | MR | MC | Profit |
|---|---|---|---|---|---|
| 4 | \$24 | \$96 | \$18 | \$10 | \$56 |
| 5 | \$22 | \$110 | \$14 | \$10 | \$60 |
| 6 | \$20 | \$120 | \$10 | \$10 | \$60 |
| 7 | \$18 | \$126 | \$6 | \$10 | \$56 |
| 8 | \$16 | \$128 | \$2 | \$10 | \$48 |

Each cartridge costs \$10 to make. No fixed cost, so ATC = MC = \$10 and profit = TR − $10Q$.

Unit 6: MR = MC = \$10, so $\Delta\pi = 0$. Optimal output 6.

Price from demand: $P(6) = \$20$, above MC = \$10.

Profit = $(20 - 10) \times 6 = \$60$.

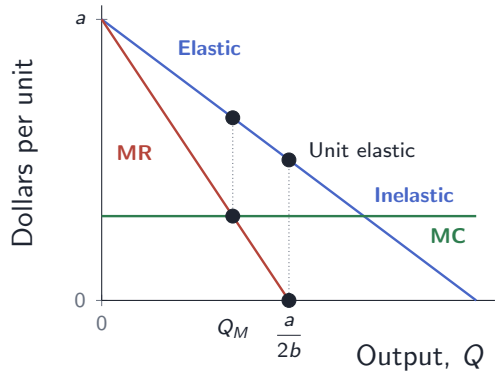
Source of the numbers: $P = 32 - 2Q$, so TR = $(32 - 2Q) \times Q$; TC = $10Q$; profit = TR − TC.

# Where on the demand curve does a monopoly produce?

Recall elasticity: on a straight-line demand curve, demand is elastic above the midpoint, unit elastic at it, and inelastic below it.

MR $= 0$ at $Q = a/(2b)$, the midpoint. To the left, MR $> 0$: elastic. To the right, MR $< 0$: inelastic.

MC is above zero, so MR $=$ MC $> 0$. So $Q_M < a/(2b)$: the elastic part.



# The cartridge seller produces on the elastic part

Demand $P(Q) = 32 - 2Q$ cuts the horizontal axis at 16, so the midpoint is $Q = 8$.

- Total revenue is highest at the midpoint: $\text{TR}(8) = 16 \times 8 = \$128$.
- The monopoly produces $Q_M = 6$, to the left of 8.
- From 6 to 7 cartridges, the price falls from \$20 to \$18 and TR rises from \$120 to \$126. A lower price raises total revenue: demand is elastic.

**Finding**

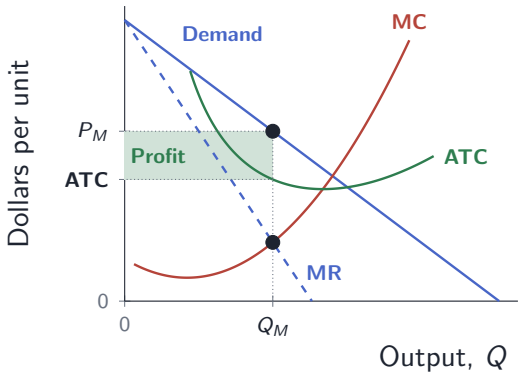
With MC $> 0$, a monopoly always produces on the elastic part of its demand curve. On the inelastic part, a higher price would raise total revenue and, with less output, lower total cost, so profit would rise.

# Profit on the graph

A general monopoly with U-shaped costs.

Choose $Q_M$ where MR = MC. Read $P_M$ from demand and ATC at $Q_M$.

$P_M >$ ATC: profit. Rectangle height $P_M -$ ATC, width $Q_M$.



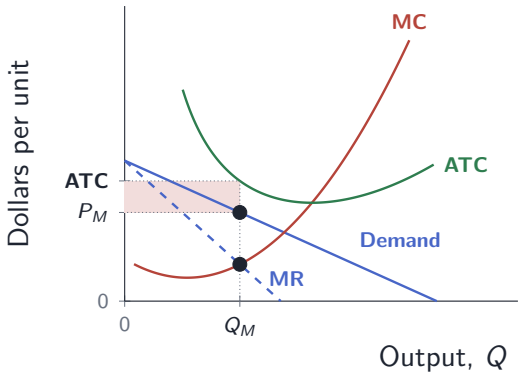
# Loss on the graph

The same costs, weaker demand.

$Q_M$ where MR = MC; $P_M$ from demand.

$P_M <$ ATC: a loss. Rectangle height ATC $- P_M$, width $Q_M$.

Being a monopoly does not guarantee a profit.



# A monopoly has no supply curve

- A supply curve gives the quantity a firm offers at each price it takes as given.
- A monopoly does not take the price as given. It chooses output and price together from MR, MC and demand.
- The same MC can go with different prices, depending on the shape of demand.

**Finding**

A competitive firm's supply curve is its MC curve above minimum AVC. A monopoly has no supply curve: its output depends on demand.

# Part 4 summary

- Profit is maximized where MR = MC: $\Delta\pi = \text{MR} - \text{MC} = 0$.
- The price comes from the demand curve at that output: $P_M = P(Q_M)$.
- Because $P >$ MR, the monopoly price is above marginal cost.
- MR = MC $> 0$, so the monopoly produces on the elastic part of demand.
- Profit $= (P_M - \text{ATC}) \times Q_M$. If $P_M <$ ATC, the monopoly makes a loss.
- A monopoly has no supply curve.

# Your turn: Check your understanding

The cost of making each extra cartridge (marginal cost) rises to \$14. There is no fixed cost.

1. The optimal (best) output is:
   A. 3  B. 5  C. 9
2. The price is:
   A. \$14  B. \$22  C. \$24
3. The profit is:
   A. \$40  B. \$70  C. \$110

| $Q$ | $P(Q)$ | TR | MR |
|---|---|---|---|
| 4 | \$24 | \$96 | \$18 |
| 5 | \$22 | \$110 | \$14 |
| 6 | \$20 | \$120 | \$10 |

# Solution: Check your understanding

1. **B.** Unit 5: MR $= \$14 =$ MC.
2. **B.** From demand: $P(5) = 32 - 10 = \$22$.
3. **A.** Profit $= (22 - 14) \times 5 = \$40$.

| $Q$ | $P(Q)$ | TR | MR |
|---|---|---|---|
| 4 | \$24 | \$96 | \$18 |
| 5 | \$22 | \$110 | \$14 |
| 6 | \$20 | \$120 | \$10 |

# Check: Part 4

Kayla's Seitan Shop is the only place within 100 miles that sells seitan Buffalo-style wings. Assuming that Kayla is a monopolist and maximizing her profit, which of the following statements is true?

A The price of Kayla's seitan wings will be less than Kayla's marginal cost.

B The price of Kayla's seitan wings will exceed Kayla's marginal cost.

C The price of Kayla's seitan wings will equal Kayla's marginal cost.

D Costs are irrelevant to Kayla because she is a monopolist.

# Check: Part 4 – answer

A profit-maximizing monopolist: which statement is true?

A The price of Kayla's seitan wings will be less than Kayla's marginal cost.

**B The price of Kayla's seitan wings will exceed Kayla's marginal cost.**

C The price of Kayla's seitan wings will equal Kayla's marginal cost.

D Costs are irrelevant to Kayla because she is a monopolist.

**Finding**

Kayla produces where MR $=$ MC and charges the price on her demand curve. For a monopoly $P >$ MR, so $P >$ MC. Costs matter: MC sets her output.

Part 5

# Monopoly versus Perfect Competition

# Two output rules

- Both firms produce where MR $=$ MC.
- Competition: MR $= \bar{P}$, so $P =$ MC.
- Monopoly: (AR $=$) $P >$ MR. Because MR $=$ MC, $P >$ MC.
- With the same demand and costs, the monopoly stops where MR, not price, meets MC: a smaller output at a higher price.

**Finding**

Monopoly: lower quantity, higher price. Competition: output where $P =$ MC.

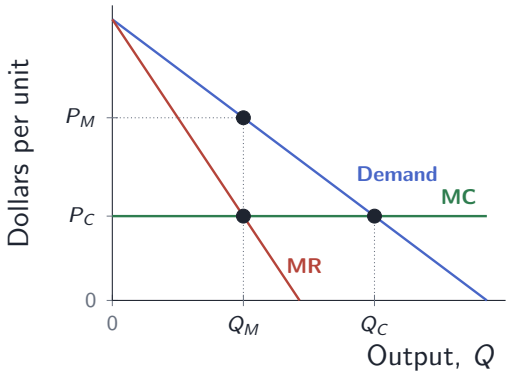
# Monopoly and competition on one graph

Same demand and constant MC.

Competition: $Q_C$ where demand meets MC; price $P_C = \text{MC}$.

Monopoly: $Q_M$ where MR meets MC; price $P_M$ from demand.

$Q_M < Q_C$ and $P_M > P_C$.



# Example: printer cartridges

One firm sells printer cartridges. Demand: $P = 32 - 2Q$. Each cartridge costs \$10 to make, so MC = \$10.

- **Monopoly, MR = MC:** the 6th cartridge raises TR from $5 \times \$22 = \$110$ to $6 \times \$20 = \$120$, so its MR is \$10, equal to MC. It sells 6 at \$20.
- **Competition, $P$ = MC:** $32 - 2Q = 10$, so $Q = 11$ and $P = \$10$.
- The monopoly sells $11 - 6 = 5$ fewer cartridges at a price $\$20 - \$10 = \$10$ higher.

| | $Q$ | $P$ | $P - \text{MC}$ |
|---|---|---|---|
| Monopoly | 6 | \$20 | \$10 |
| Competition | 11 | \$10 | \$0 |

# The deadweight loss of monopoly

**Definition: Deadweight loss**

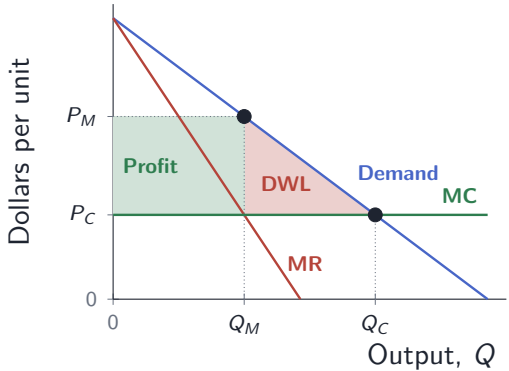
The fall in total surplus when output is below the efficient level.

- The efficient output is where $P = \text{MC}$: buyers' value of the last unit equals its cost.
- The monopoly stops at $Q_M < Q_C$. For each unit between them, buyers' value (demand) is above its cost (MC), but the unit is not made.
- Cartridges: the triangle between demand and MC, $\frac{1}{2} \times (11 - 6) \times (20 - 10) = \$25$.

# The deadweight loss on the graph

The triangle between demand and MC, from $Q_M$ to $Q_C$, is the deadweight loss.

With no fixed cost, ATC = MC, so the profit rectangle is $(P_M - \text{MC}) \times Q_M$. It is not lost: it moves from buyers to the firm.




**Finding**

Monopoly profit is a transfer from buyers to the firm. The deadweight loss is surplus that no one gets: the social cost of monopoly.

# The two market structures

| | Perfect competition | Monopoly |
|---|---|---|
| Number of firms | Many | One |
| Product | Identical | No close substitutes |
| Entry | Free | Blocked by barriers |
| Demand facing the firm | Horizontal at $\bar{P}$ | Market demand, $P(Q)$ |
| Price and MR | $P = \text{MR}$ | $P > \text{MR}$ |
| Output rule | $P = \text{MC}$ | $\text{MR} = \text{MC}$, $P > \text{MC}$ |
| Long-run profit | Zero | Can stay positive |
| Deadweight loss | None | Yes |

# Part 5 summary

- Competition produces where $P = \text{MC}$; monopoly where $\text{MR} = \text{MC}$, with $P > \text{MC}$.
- With the same demand and costs, monopoly output is lower and its price higher.
- The units between $Q_M$ and $Q_C$ are worth more to buyers than they cost: their lost surplus is the deadweight loss, $\frac{1}{2} \times (Q_C - Q_M) \times (P_M - \text{MC})$ for straight-line demand and constant MC.
- Monopoly profit is a transfer, not a deadweight loss.

# Your turn: Check your understanding

Printer cartridges. Demand: $P = 32 - 2Q$. Each cartridge costs \$14 to make, so MC = \$14. The monopoly sells 5 cartridges at \$22 each.

1. The competitive output, where $P = \text{MC}$, is:
   A. 5  B. 9  C. 14
2. The monopoly price is above MC by:
   A. \$0  B. \$8  C. \$22
3. The deadweight loss, $\frac{1}{2} \times (Q_C - Q_M) \times (P_M - \text{MC})$, is:
   A. \$16  B. \$32  C. \$40

| | $Q$ | $P$ |
|---|---|---|
| Monopoly | 5 | \$22 |
| Competition | ? | \$14 |

# Solution: Check your understanding

1. **B.** $P = \text{MC}$: $32 - 2Q = 14$, so $Q = 9$.
2. **B.** $22 - 14 = \$8$.
3. **A.** Triangle:
   $\frac{1}{2} \times (9 - 5) \times (22 - 14) = \frac{1}{2} \times 4 \times 8 = \$16$.

| | $Q$ | $P$ |
|---|---|---|
| Monopoly | 5 | \$22 |
| Competition | 9 | \$14 |

# Check: Part 5

Monopolies are socially inefficient because the price they charge is

A equal to marginal revenue.

B above marginal cost.

C equal to demand.

D above demand.

# Check: Part 5 – answer

Monopolies are socially inefficient because the price they charge is

A equal to marginal revenue.

**B above marginal cost.**

C equal to demand.

D above demand.

**Finding**

With $P >$ MC, some buyers value extra units above their cost, but the monopoly does not produce them. That lost surplus is the deadweight loss. Efficiency needs $P =$ MC.

Part 6

# Price Discrimination and Public Policy

# What is price discrimination?

**Definition: Price discrimination**

Selling the same good at different prices to different buyers, when the difference in price is not due to a difference in cost.

**Examples**

- **Airline seats:** two passengers on the same flight pay very different fares, depending on when they book.
- **Cinema tickets:** people over 65 pay less for the same film on a Thursday afternoon.

# What a firm needs to price discriminate

1. **Market power.** A competitive firm takes the price as given, so it cannot set different prices.
2. **A way to segment the market.** The firm must sort buyers into groups that will pay different amounts (by booking date, by age).
3. **No resale.** Buyers who pay the low price must not be able to resell to buyers who would pay the high price.

**Finding**

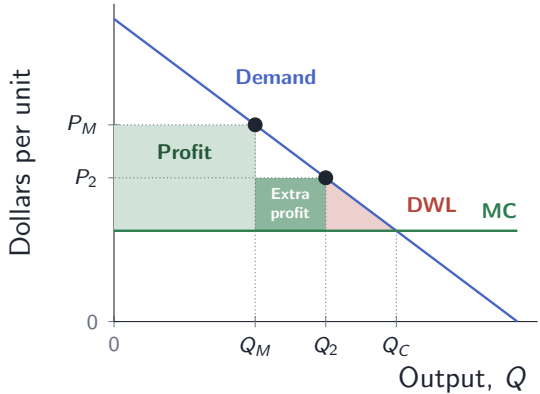
The airline can do all three: it is one of few sellers on a route, booking date sorts travelers, and tickets carry the passenger's name.

# Two prices instead of one

With one price, the monopoly sells $Q_M$ at $P_M$ (Part 4). Units from $Q_M$ to $Q_C$ are lost: the deadweight loss.

Now add a lower price, $P_2$, only for buyers who will not pay $P_M$.

Output rises to $Q_2$. Profit rises by the dark rectangle. The deadweight loss shrinks.




**Finding**

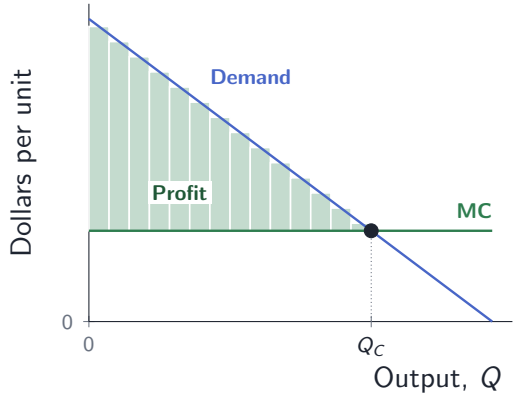
Price discrimination raises profit and output, and cuts the deadweight loss.

# Perfect price discrimination

Now charge every buyer exactly the most that buyer will pay. Each strip is one buyer.

One more sale does not lower the price to anyone else, so MR $= P$.

The firm sells until $P =$ MC: output $Q_C$, the competitive output.




**Finding**

No deadweight loss: every unit worth more than MC is sold. All surplus is profit.

# Public policy toward monopolies

1. **Antitrust law:** block mergers and break up firms that reduce competition.
2. **Regulation:** set the price the monopoly may charge.
3. **Public ownership:** the government runs the monopoly.
4. **Do nothing:** when the cure costs more than the deadweight loss.

**Finding**

Natural monopoly is the hard case: breaking it up would raise costs, so the usual choice is regulation or public ownership.

# Regulating a natural monopoly

The town water network: $\text{TC} = 120 + 3Q$, $\text{MC} = \$3$. At 60 connections, $\text{ATC} = \$5$.

- **Price = MC = \$3:** efficient, but $(5 - 3) \times 60 = \$120$ loss, the whole fixed cost. The firm needs a subsidy or it leaves.
- **Price = ATC = \$5:** revenue $5 \times 60 = \$300$ covers $\text{TC} = \$300$: zero profit. But $P > \text{MC}$: homes take fewer water connections at \$5 than at \$3, so some deadweight loss remains.

**Finding**

For a natural monopoly ATC is above MC, so a price equal to MC always makes a loss.

# Part 6 summary

- Price discrimination: the same good at different prices to different buyers.
- It needs market power, a way to segment the market, and no resale.
- A second, lower price raises profit and output, and cuts the deadweight loss.
- Perfect price discrimination: output $Q_C$, no deadweight loss, all surplus to the firm.
- Policy options: antitrust, regulation, public ownership, or doing nothing.
- Regulating a natural monopoly at $P = \text{MC}$ causes a loss; at $P = \text{ATC}$ it breaks even.

# Your turn: Check your understanding

1. Which is price discrimination?
   A. A larger TV costs more than a smaller one.
   B. A cinema charges people over 65 less for the same film.
   C. A store raises its price after its costs rise.
2. An airline can charge two fares on one flight only if:
   A. tickets cannot be resold. B. it has many rivals. C. every seat costs the same.
3. With perfect price discrimination, the deadweight loss is:
   A. larger than with one price. B. zero. C. unchanged.

# Solution: Check your understanding

1. **B.** Same film, same cost, different prices. In A the goods differ; in C the price follows cost.
2. **A.** If a cheap ticket could be resold, the high-fare buyers would buy it.
3. **B.** Every unit worth more than MC is sold: output reaches $Q_C$.

# Check: Part 6

Price discrimination

A is illegal in the United States and Europe.

B can occur in both perfectly competitive and monopoly markets.

C is illogical because it does not maximize profits.

D can maximize profits if the seller can prevent the resale of goods between customers.

# Check: Part 6 – answer

Price discrimination

A is illegal in the United States and Europe.

B can occur in both perfectly competitive and monopoly markets.

C is illogical because it does not maximize profits.

**D can maximize profits if the seller can prevent the resale of goods between customers.**

**Finding**

Charging each group what it will pay raises profit, as on the two-price graph. It needs market power (so not competition) and no resale between buyers.

# Which rule answers which question

| Question | Rule |
|---|---|
| What demand does a monopoly face? | The market demand curve: $P = P(Q)$. |
| Why does a monopoly exist? | Barriers to entry: a key resource, government rights, or natural monopoly. |
| How do P and MR compare? | MR $<$ $P$: the price effect. |
| How much should it produce? | MR $=$ MC. Increase output if MR $>$ MC; cut it if MC $>$ MR. |
| What price does it charge? | The price on the demand curve at that output: $P >$ MC. |
| Where on demand does it produce? | The elastic part: MR $=$ MC $> 0$. |
| Does it make a profit? | $\pi = (P - \text{ATC}) \times Q$; a loss if $P <$ ATC. |
| How does it compare with competition? | Lower output, higher price, a deadweight loss. |